

Jay Barad

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AI Engineer and Data Science undergraduate with hands-on experience designing and shipping production agentic systems – including LangGraph-orchestrated multi-step agents, MCP-based retrieval tools, and RAG pipelines combining keyword, semantic, and lexical search. Experienced building LLM-powered applications end-to-end: from tool design and prompt engineering to human-in-the-loop workflows, LLM observability with Langfuse, and deployment on FastAPI backends with async infrastructure. Strong foundation in Python, SQL, and deep learning (PyTorch, TensorFlow, OpenCV), with a track record of collaborating cross-functionally to turn research-grade ideas into reliable, production-ready systems.

EDUCATION

Mukesh Patel School of Technology Management and Engineering

Aug. 2022 – Present

Bachelor of Technology in Data Science

Mumbai

- **GPA:** 4.00/4.00
- **CGPA:** 3.79/4.00

Lilavatibai Podar High School

2020 – 2022

ISC

Mumbai

- **12th Grade Score:** 91.0

Lokhandwala Foundation School

2010 – 2020

ICSE

Mumbai

- **10th Grade Score:** 93.4

SKILLS

Computer Vision & Deep Learning : PyTorch, TensorFlow, Keras, OpenCV, Sci-kit Learn, CNN-based classification, Matplotlib, Seaborn

Programming & Data : Python, SQL (PostgreSQL, MySQL, Oracle), Pandas, Numpy, Snowflake, Clickhouse, Dbeaver, Microsoft Excel

Software Engineering & MLOps : Git, FastAPI, Docker, Pyinstaller, Pytest, Selenium, Streamlit, MLFlow, Apache Airflow, AWS, Amazon Bedrock, MongoDB, Neo4j, Langfuse, Langchain, React.js, Django, Tailwind, Bootstrap

EXPERIENCE

InSync Analytics

Sept. 2025 – June 2026

Data Science Intern

Mumbai

- Engineered a real-time analytics pipeline (NER + AI-based retrieval) as part of the team's production AI stack, cutting discovery-to-alert latency to under 60 seconds and debugging complex issues to ensure reliability at scale.
- Built and deployed an MCP-based retrieval tool combining keyword, semantic, and lexical search (RAG) to surface relevant insights from large unstructured datasets.
- Improved robustness and operational efficiency of an internal data-cleaning tool by packaging Streamlit apps as standalone executables and building scalable, low-overhead FastAPI backend endpoints.

DataEngite

April. 2025 – Sept. 2025

Data Science Intern

Mumbai

- Built a dynamic Excel data-cleaning solution to standardize inconsistent file structures for efficient, accurate data ingestion.
- Developed a data governance engine using Small Language Models to semantically map columns to master data, improving data quality and consistency for downstream analysis.

Collegestreet.tech

Feb 2025 – June. 2025

Data Science Intern

Mumbai

- Built AI pipelines that synthesized news articles, quarterly reports, and conference call transcripts into structured financial reports, translating raw data into decision-ready insights.

PROJECTS

Unattended Quotation Agent

- Built a fully unattended, multi-tenant agentic quotation system for a distribution business, orchestrated via a LangGraph with a checkpointing for crash-safe durability.
- Engineered a layered SQL-first product retrieval engine (exact code match, boundary-anchored alias suffix, PostgreSQL FTS and trigram fallback) re-ranked with RapidFuzz, achieving robust matching against typos, abbreviations, and partial SKU codes.
- Designed a human-in-the-loop gate and Built an in-process SSE streaming endpoint (replay of checkpointed history + live in-process pub/sub broadcaster) allowing operators to attach mid-run without racing or re-invoking the graph.

Brain Dump Sanctuary

- Built a Streamlit application that captures free-form thoughts, semantically embeds and clusters them using HDBSCAN + UMAP, and auto-labels each cluster using an LLM chain persisted in a Neo4j knowledge graph with cosine-similarity relationships visualized interactively.
- Orchestrated a parallel multi-agent pipeline using LangGraph that concurrently generates Socratic questions, multi-perspective analysis, and web-synthesized research for each thought, with results cached back to Neo4j.

Face Recognition Photo Filtering Solution

- Developed and deployed a computer vision pipeline to filter photos based on users' facial data using FaceNet embeddings.
- Benchmarked multiple face detection/recognition architectures – MTCNN, ResNet50, and a custom CNN – to select the best-performing approach; Classified via SVM and deployed via Streamlit.

AvianIntelligence

- Built a bird species recognition system operating on two modalities – photo and audio (bird calls) – to handle diverse real-world identification scenarios.
- Used a CNN (ResNet50) for image-based classification, Librosa for audio feature extraction, and Random Forest for audio-based classification.

Bird Strike Analysis

- Built an interactive Power BI dashboard on a large web-scraped incident dataset to surface trends and support data-driven decisions.
- Trained a Random Forest classifier to predict damage severity across diverse real-world conditions (airport, time of year, species, flight phase).
- Used an XGBoost Regressor to estimate repair costs, adding a quantifiable business-impact layer to the analysis.

Agri-Horticulture commodity prices prediction and forecasting, Smart India Hackathon

- Built a prototype forecasting model to determine optimal selling price, benchmarking multiple deep learning architectures (Dense, Conv1D, LSTM) to identify the best-performing model.

CERTIFICATIONS

- Neural Networks and Deep Learning DeepLearning.AI (Coursera)
- Convolutional Neural Networks DeepLearning.AI (Coursera)
- Improving Neural Networks: Hyperparameter Tuning, Regularization and Optimization DeepLearning.AI (Coursera)
- Structuring Machine Learning Projects DeepLearning.AI (Coursera)
- Sequence Models DeepLearning.AI (Coursera)
- SQL 50 Badge (Leetcode)
- Data Modeling in Power BI (Coursera)
- Share Data Through the Art of Visualization Google (Coursera)
- Google Data Analytics Capstone: A Complete Case Study Google (Coursera)
- Data Analysis with R Programming Google (Coursera)
- Introduction to Cloud Computing IBM (Coursera)